Containerization-and-DevOps

🐳 Docker Daemon Configuration & TCP Access – Class Task

🛠️ Technologies Used

- Docker
- WSL (Windows Subsystem for Linux)
- Linux Commands

⚙️ Steps Performed

1️⃣ Edit Docker Daemon Configuration

```
sudo nano /etc/docker/daemon.json
```

Updated Configuration:

```
{
  "hosts": [
    "unix:///var/run/docker.sock",
    "tcp://0.0.0.0:2375"
  ]
}
```

2️⃣ Reload Docker Configuration

```
sudo systemctl daemon-reexec
```

3️⃣ Restart Docker Service

```
sudo systemctl restart docker
```

4️⃣ Check Docker Service Status

```
sudo service docker status
```

5️⃣ Stop and Start Docker Service

```
sudo service docker stop
sudo service docker start
```

6️⃣ Kill and Restart Docker Daemon (Manual)

```
sudo pkill dockerd
sudo dockerd &
```

7️⃣ Verify Docker Command

```
docker
```

8️⃣ Check Port Binding (2375)

```
ss -lntp | grep 2375
```

⚠️ Notes

- Docker daemon was configured to listen on both:
  - Unix socket
  - TCP port `2375`
- Port `2375` allows remote API access (insecure if not protected)

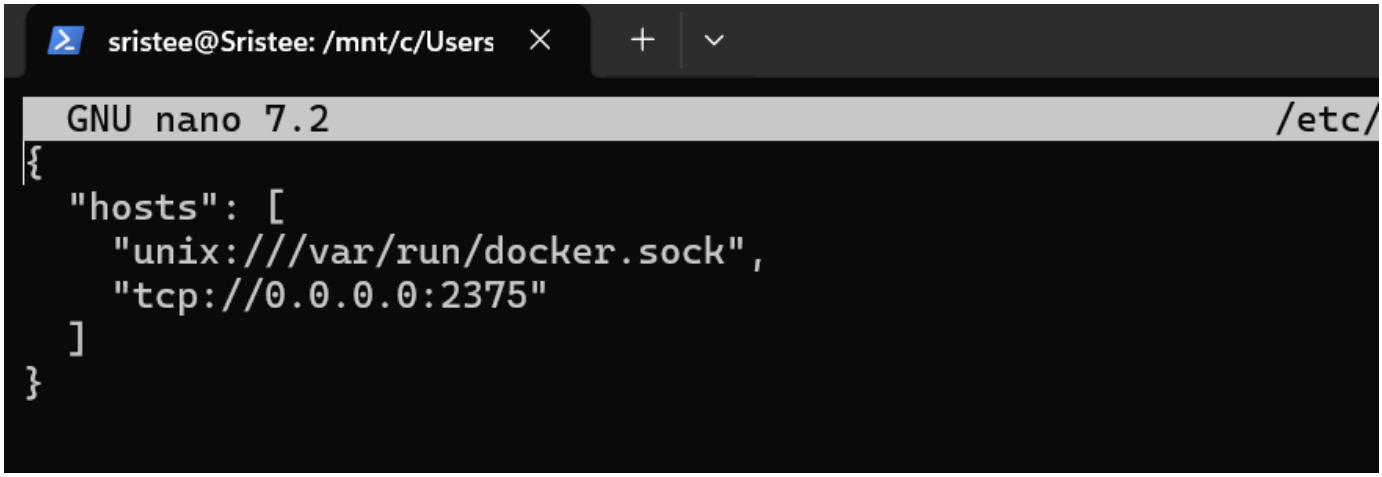
- Restarting Docker is required after configuration changes
- Manual daemon restart can be used for troubleshooting

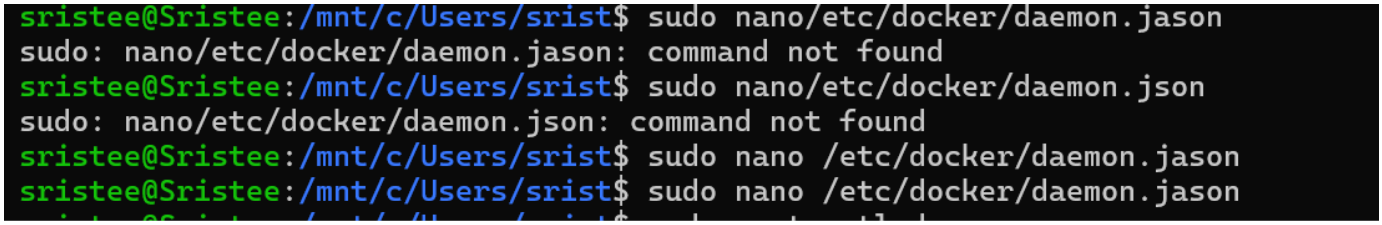
✅ Outcome

- Successfully modified Docker daemon configuration
- Enabled TCP access for Docker API
- Restarted and verified Docker service
- Confirmed port binding for external communication

📸 Screenshots

> all screenshots below:





```
sristee@Sristee:/mnt/c/Users/srist$ sudo service docker status
```

```
Feb 04 06:01:36 Sristee dockerd[2387]: time="2026-02-04T06:01:36.896884338Z" level=info msg="Loading containers: done."
Feb 04 06:01:36 Sristee dockerd[2387]: time="2026-02-04T06:01:36.926340707Z" level=info msg="Docker daemon" commit="28.2.2-0ubuntu1~>
Feb 04 06:01:36 Sristee dockerd[2387]: time="2026-02-04T06:01:36.926462777Z" level=info msg="Initializing buildkit"
```

```
sristee@Sristee:/mnt/c/Users/srist$ ss -lntp | grep 2375
```